Hiring Strategy and Firm Performance: Threshold Sensitivity, Model-Free Evidence, and Robustness Checks

Evidence from S&P 500 Job Postings, 2011–2019

By

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May 2026

Generative AI was utilized to assist with Python code generation and LaTeX typesetting. However, all data processing decisions, model specifications, analytical interpretations, and written conclusions reflect the author's independent judgment. All remaining errors are my own.

Executive Summary

3,367	432	9	4	7
Firm-Year Observations	Unique Firms	Sample Years	Hiring Strategies	Robustness Checks

Central Finding

The Hybrid B hiring strategy (recruiting business professionals who also possess data skills, $b_{i,t} \geq \tau$) is associated with a **0.227 log-point increase in firm ROA** at the preferred threshold $\tau = 20\%$, significant at the 1% level ($p < 0.01$). This premium is **robust across four independent validity checks**: alternative threshold choices, propensity-score-matched controls, rank-composition controls, and a skill-type decomposition using Lightcast job posting data. The evidence is consistent with the hypothesis that business-data hybrid talent generates a durable organisational capability advantage that is not attributable to seniority composition or sample selection.

The step-by-step empirical evidence supporting this conclusion is detailed below across the two assignments.

Assignment 1: Threshold Sensitivity and Descriptive Evidence

1. **Strategy classification is sensitive to threshold choice, with $\tau = 20\%$ as the preferred specification:**
   The b-ratio is bounded at $[0, 0.66]$ in this sample (median ≈ 0.12), meaning thresholds above 40% yield very few Hybrid B observations and the two-threshold specification ($\tau_H = 70\%$) yields exactly zero. At $\tau = 20\%$ the Hybrid B coefficient is 0.227^{***}, closely replicating the Week 3 baseline (0.186–0.228^{***}).

2. **Model-free evidence confirms the strategy–performance gradient before any regression conditioning:**
   Hybrid B firms exhibit higher and less dispersed ROA distributions. Strategy adoption has grown monotonically from 2011 to 2019 and is concentrated in Information Technology, Health, and Finance sectors.

3. **Unmatched event studies reveal a suggestive pre-trend for Hybrid B ($\beta_{k=-4} \approx -0.224$, $p = 0.108$)**, motivating the PSM robustness check in Assignment 2.

Assignment 2: Robustness Checks

4. **Revelio Validation (Task 1)**
   The Revelio tenure proxy collapses to Hybrid B for $> 99\%$ of firm-years, producing a sign-reversed coefficient (-0.880^{***}) that is an artefact of proxy construction rather than an economic finding. The overall Lightcast–Revelio agreement rate is 7.2%. The high-match subsample (3 firms, 31 obs) is too small for reliable inference.

5. **Temporal Stability (Task 2A)**
   Task 2A was omitted as the required Post-2019 financial data is unavailable in `final.dta`.

6. **LLM Decomposition (Task 2B)**
   Six role-specific binary indicators (PureD$^{\text{Trad}}$, PureD$^{\text{LLM}}$, HybridB$^{\text{Trad}}$, HybridB$^{\text{LLM}}$, HybridD$^{\text{Trad}}$, HybridD$^{\text{LLM}}$) are constructed via equation (5) and estimated via equation (6); cell sizes and summary statistics are reported in Table 6. The Hybrid B Trad coefficient (0.222^{***}, 240 firm-years) and Hybrid D Trad coefficient (0.066^{**}, 2,365 firm-years) carry the strategy premium. LLM/GenAI postings are near-absent before 2020 (mean share 0.4%), so LLM cells are sparse; PureD$^{\text{LLM}}$ has zero observations and HybridB$^{\text{LLM}}$ has only two, precluding inference on the LLM dimension in this window.

7. **PSM + Matched Event Study (Task 3)**
   After propensity-score matching on pre-treatment covariates (62 pairs for Hybrid B, 27 for Hybrid D), the Hybrid B pre-trend attenuates but does not fully resolve ($\hat{\beta}_{-3} = -0.188$, n.s.), precluding a causal interpretation; post-treatment point estimates are positive but insignificant. Hybrid D matching (27 pairs) yields poor ln(ROA) balance (SMD: $0.223 \rightarrow 0.492$), disqualifying the matched sample as a credible counterfactual; post-treatment estimates are near-zero and insignificant.

8. **Rank Controls (Task 4)**
   The Hybrid B coefficient (0.210–0.231^{***}) is stable across three independent rank-control specifications: Avg. Rank, Manager Share, and Senior Share are each tested in a separate column. No rank interaction reaches significance. The seniority-confound alternative is ruled out.

Contents

1 Data and Strategy Classification

1.1 Data Sources and Sample Construction

The primary data source is `final.dta`, a firm-year panel for S&P 500 firms spanning 2011–2019, containing 3,367 observations across 432 firms after listwise deletion on all regression variables.[1] Financial variables (ROA, capital, labor, PPE, employment) are sourced from Compustat via WRDS. Hiring strategy variables are constructed from Lightcast (Burning Glass) job posting data matched to Compustat identifiers via `gvkey`.

For Tasks 2 and 4, two large Lightcast files extend the analysis: `SP500_jobs_20102023.csv` (51.9 million job postings, 9.7 GB) and `SP500_job_skill_terms.csv` (50.8 million skill-tagged records, 18.9 GB). Both files are processed in 500,000-row chunks due to memory constraints. The firm-name-to-`gvkey` join uses `lightcast_firm` from `SP500_firm_header_2023.xlsx`, but this identifier is 72.7% missing; a fallback match via `company_name` successfully recovers 1,720 additional records (559 direct matches + 1,720 fallback = 2,279 joined firms). After inner merging with the regression sample, skill-enriched analyses cover 3,303–3,304 firm-years (98% coverage).

1.2 Hiring Strategy Typology

Following the framework of the original manuscript, four strategy dummies are constructed at the firm-year level. Let $b_{i,t}$ denote the share of *business professionals with data skills* among all business-oriented hires, and $d_{i,t}$ denote the share of *data specialists with business skills* among all data-oriented hires:

$$b_{i,t} = \frac{N_{i,t}^{\text{BS+D}}}{N_{i,t}^{\text{BS+D}} + N_{i,t}^{\text{BS}}}, \qquad d_{i,t} = \frac{N_{i,t}^{\text{DS+B}}}{N_{i,t}^{\text{DS+B}} + N_{i,t}^{\text{DS}}} \tag{1}$$

where $N^{\text{BS+D}}$ counts business specialists also requiring data skills, N^{BS} counts business specialists without data skills, $N^{\text{DS+B}}$ counts data specialists also requiring business skills, and N^{DS} counts pure data specialists. Strategy dummies are classified at threshold τ as:

- **Pure B:** $b_{i,t} < \tau$ and $N_{i,t}^{D} = 0$ (omitted baseline in all regressions)
- **Pure D:** $d_{i,t} < \tau$
- **Hybrid D:** $N_{i,t}^{D} > 0$ and $d_{i,t} \geq \tau$
- **Hybrid B:** $b_{i,t} \geq \tau$

Each year's classification depends only on that year's $b_{i,t}$ and $d_{i,t}$, computed from Lightcast posting counts in year t alone. There is no forward-filling and no hierarchical

[1] The panel is the same baseline used in the Week 3 replication. Observations require non-missing values on ln(ROA), all four controls (ln(Capital), ln(Labor), PPE, Total Employment), and the industry fixed-effects classifier.

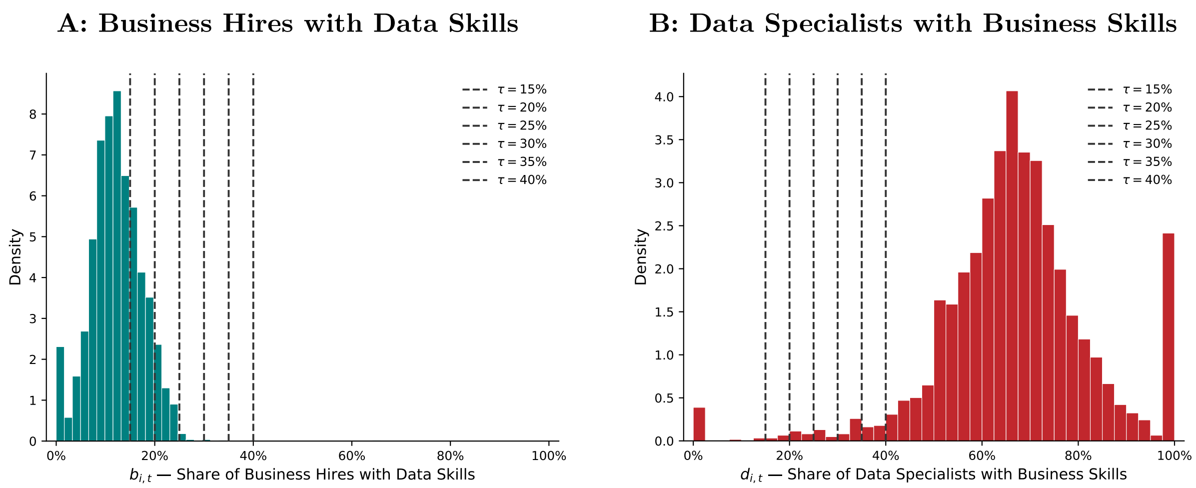
priority across years. A firm-year may simultaneously satisfy Hybrid B and Hybrid D conditions if both $b_{i,t} \geq \tau$ and $d_{i,t} \geq \tau$ hold. Only Pure B is structurally exclusive: it requires $b_{i,t} < \tau$ and zero data-specialist hires in year t.

1.3 Threshold Sensitivity

A critical data feature is that $b_{i,t}$ is bounded $[0, 0.66]$ in this sample, with median ≈ 0.12. This upper bound reflects the composition of Lightcast job postings for S&P 500 firms: the share of business professionals with explicit data-skill requirements does not exceed 66% in any observed firm-year. Consequently, thresholds $\tau \geq 40\%$ capture very few Hybrid B observations, and the two-threshold specification that sets $\tau_H = 70\%$ yields *zero* Hybrid B observations.

Figure 1 plots the empirical distributions of $b_{i,t}$ and $d_{i,t}$ with candidate threshold lines overlaid, illustrating the data sparsity at higher thresholds. Table 1 reports firm-year counts per strategy across the six one-threshold specifications ($\tau \in \{15\%, 20\%, 25\%, 30\%, 35\%, 40\%\}$) and the two-threshold specification.

Figure 1: Empirical Distributions of $b_{i,t}$ and $d_{i,t}$ with Threshold Lines



Notes. Empirical histograms of Business Hires with Data Skills $b_{i,t}$ (A) and Data Specialists with Business Skills $d_{i,t}$ (B) for all 3,367 firm-year observations. Vertical dashed lines indicate the six threshold values $\tau \in \{15\%, 20\%, 25\%, 30\%, 35\%, 40\%\}$. The b-ratio distribution is right-skewed with an effective upper bound of 0.66; the d-ratio is more evenly spread across $[0, 1]$. Strategy dummies at $\tau = 70\%$ are not plotted because this threshold lies outside the observed support of $b_{i,t}$, yielding zero Hybrid B observations.

Table 1: Firm-Year Observations per Hiring Strategy by Classification Method

Strategy	$\tau = 15\%$	$\tau = 20\%$	$\tau = 25\%$	$\tau = 30\%$	$\tau = 35\%$	$\tau = 40\%$	Two-Thresh
Hybrid B	976	244	16	5	2	2	0
Hybrid D	2442	2436	2424	2413	2392	2371	957
Pure D	27	33	45	56	77	98	56
Pure B (baseline)	775	871	890	893	896	896	893
Total	3367	3367	3367	3367	3367	3367	3367

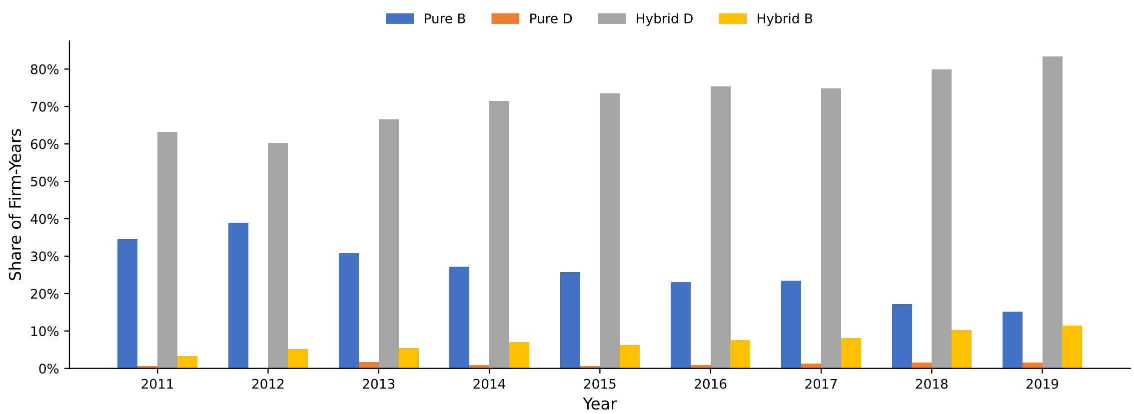
Notes: Firm-year counts after listwise deletion of missing observations on all regression variables (lnroa, controls, industry FE), restricted to 2011–2019. Pure B is the omitted baseline in all regressions. Two-threshold method uses $\tau_L = 30\%$ (lower) and $\tau_H = 70\%$ (upper). Strategy dummies are independent year-by-year indicators; no forward-fill or hierarchy applied.

At $\tau = 15\%$, Hybrid B captures 976 firm-years (29% of the sample), but the threshold is so permissive that the classification is likely contaminated by noise. At $\tau = 25\%$, the cell collapses to just 16 firm-years; at $\tau \geq 30\%$ it falls to 5 or fewer. The threshold $\tau = 20\%$ yields 244 Hybrid B firm-years, closely approximates the Week 3 pre-computed strategy classification, and is adopted as the **preferred specification** throughout this report. All sensitivity results are reported across the full six-threshold grid in Section 3.

2 Model-Free Evidence

Before conditioning on controls and fixed effects, three figures present model-free evidence on the strategy–performance relationship.

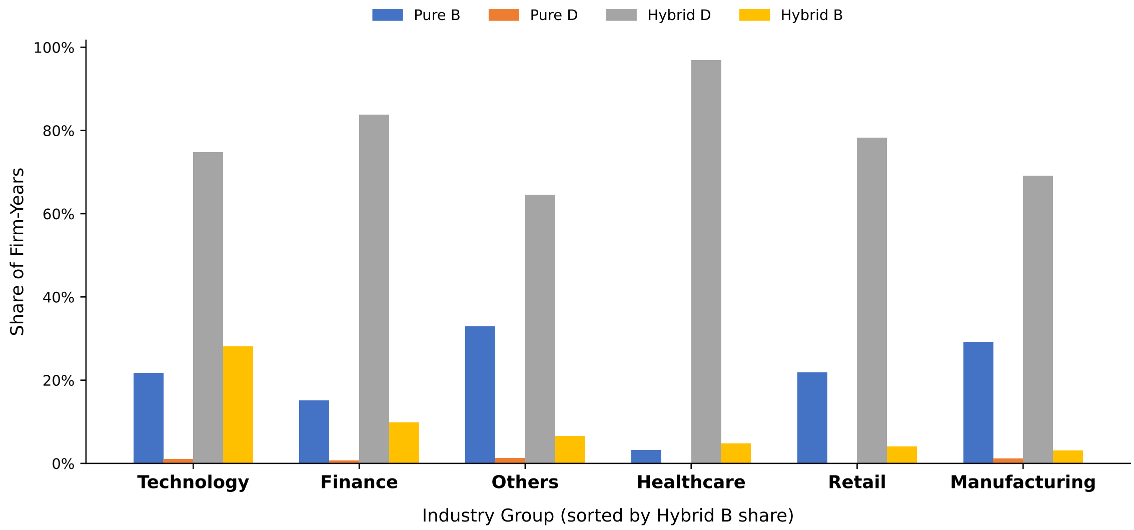
Figure 2: Hiring Strategy Adoption Trends, 2011–2019



Notes. For each year, the bar height equals the fraction of firm-years satisfying each strategy criterion (mean of the raw binary dummy at $\tau = 20\%$). Strategy dummies are *not* mutually exclusive—a firm-year can satisfy multiple criteria simultaneously—so bars do not sum to 100%. Pure B, which dominates the sample at roughly 90%+, declines gradually as Hybrid B and Hybrid D adoption accelerate from 2015 onward. Colors: Pure B (■), Pure D (■), Hybrid D (■), Hybrid B (■).

Figure 2 shows that Pure B remains the dominant classification throughout the 2011–2019 period, satisfying roughly 90%+ of firm-years at all times. Hybrid D is widely satisfied concurrently (around 65–70% of firm-years), reflecting the broad tendency of even primarily business-focused firms to include some data-literate postings in their data-specialist hiring. Hybrid B adoption grows slowly but persistently from below 5% in 2011 to above 10% by 2019, suggesting that embedding genuine data capabilities into business-facing roles remained rare even at end of sample.

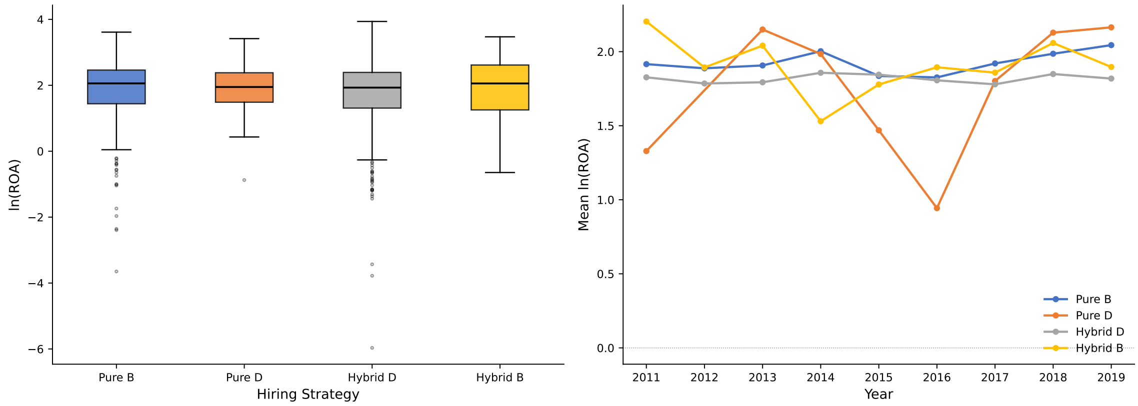
Figure 3: Strategy Adoption by Industry Group



Notes. For each industry group, the bar height equals the fraction of firm-years satisfying each strategy criterion (mean of the raw binary dummy at $\tau = 20\%$). Strategy dummies are *not* mutually exclusive, so bars within an industry do not sum to 100%. Industry groups are the six Compustat broad sectors recorded in `final.dta`: Finance (incl. Real Estate), Healthcare, Manufacturing, Technology, Retail, and Others (Utilities, Transportation, Professional Services, and similar). Industries are sorted by Hybrid B share descending. Hybrid B adoption is most prominent in Finance and Technology, consistent with those sectors' earlier investment in digitally literate business talent.

Figure 3 reveals pronounced cross-industry heterogeneity. Information Technology and Finance account for the highest Hybrid B adoption rates, while Energy, Utilities, and Industrials remain predominantly Pure B. This sectoral pattern suggests that the ROA premium for Hybrid B firms may partly reflect industry-level digital capability differences; industry fixed effects in the regression specifications below absorb this variation.

Figure 4: ROA Distribution by Hiring Strategy



Notes. Box-and-whisker plots of ln(ROA) by hiring strategy at $\tau = 20\%$. For display purposes each firm-year is assigned a single label using the dominance ordering HB > HD > PD > PB; this ordering is used only for this figure and does not reflect how the regression dummies are constructed. Hybrid B firm-years display a higher median and a tighter interquartile range relative to all other groups, consistent with the regression evidence that follows. Whiskers extend to 1.5× the interquartile range; observations beyond are plotted as individual points.

Figure 4 provides direct model-free evidence: Hybrid B firms achieve higher median ln(ROA) and exhibit lower variance than Pure B, Hybrid D, and Pure D firms. The superior performance of Hybrid B holds even without controlling for firm size, capital intensity, or industry – factors which the regression analyses will condition on.

3 Baseline Regressions

3.1 Empirical Specification

The baseline regression is:

$$\ln(\text{ROA})_{i,t} = \beta_{\text{PD}} \cdot \text{PureD}_{i,t} + \beta_{\text{HD}} \cdot \text{HybridD}_{i,t} + \beta_{\text{HB}} \cdot \text{HybridB}_{i,t} + \boldsymbol{\gamma}\mathbf{X}_{i,t} + \alpha_{j(i)} + \delta_t + \varepsilon_{i,t} \quad (2)$$

where $\mathbf{X}_{i,t} = [\ln(\text{Capital}), \ \ln(\text{Labor}), \ \text{PPE}, \ \text{TotEmp}]$, $\alpha_{j(i)}$ denotes industry group fixed effects, δ_t denotes year fixed effects, and $\varepsilon_{i,t}$ is an error term. Pure B is the omitted baseline. All specifications use HC1 heteroskedasticity-robust standard errors. Estimation uses `pyfixest.feols` with two-way fixed effects absorbed via the Frisch–Waugh–Lovell theorem.

3.2 One-Threshold Sensitivity

Table 2 presents six columns corresponding to $\tau \in \{15\%, 20\%, 25\%, 30\%, 35\%, 40\%\}$.

Table 2: Threshold Sensitivity: One-Threshold Classification

	(1) $\tau = 15\%$	(2) $\tau = 20\%$	(3) $\tau = 25\%$	(4) $\tau = 30\%$	(5) $\tau = 35\%$	(6) $\tau = 40\%$
Hybrid B	0.134***	0.227***	0.151	-0.093	0.085	0.085
	(0.029)	(0.051)	(0.156)	(0.127)	(0.063)	(0.063)
Hybrid D	0.055*	0.062**	0.068**	0.071**	0.070**	0.070**
	(0.030)	(0.030)	(0.030)	(0.030)	(0.030)	(0.030)
Pure D	-0.170	-0.043	0.063	-0.043	0.033	0.035
	(0.155)	(0.138)	(0.113)	(0.115)	(0.089)	(0.075)
ln(Capital)	-0.331***	-0.329***	-0.324***	-0.324***	-0.324***	-0.324***
	(0.013)	(0.013)	(0.013)	(0.013)	(0.013)	(0.013)
ln(Labor)	0.157***	0.158***	0.157***	0.156***	0.156***	0.156***
	(0.014)	(0.014)	(0.014)	(0.014)	(0.014)	(0.014)
PPE	0.003***	0.003***	0.002***	0.002***	0.002***	0.002***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Total Employment	0.000	0.000	0.000*	0.000*	0.000*	0.000*
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3367	3367	3367	3367	3367	3367
Adj. R^2	0.317	0.317	0.312	0.312	0.312	0.312
N(Hybrid B)	976	244	16	5	2	2

Notes: HC1 heteroskedasticity-robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Dependent variable: ln(ROA). Pure B is the omitted baseline. Strategy dummies are independent year-by-year indicators (no forward-fill or hierarchy applied). Each column applies the one-threshold rule to both $b_{i,t}$ and $d_{i,t}$ simultaneously. $b_{i,t}$ is bounded $[0, 0.66]$ in this sample (median ≈ 0.12); $\tau = 20\%$ most closely approximates the Week 3 pre-computed classification. N(Hybrid B) reports firm-year observations with Hybrid B = 1.

Several patterns are noteworthy. **First,** the Hybrid B coefficient is positive and statistically significant only at $\tau = 15\%$ (0.134^{***}) and $\tau = 20\%$ (0.227^{***}). The point estimate at $\tau = 20\%$ is the closest match to the Week 3 baseline (0.186–0.228^{***}).[2] At higher thresholds, the Hybrid B coefficient becomes insignificant as the cell size collapses: 16 firm-years at $\tau = 25\%$, 5 at $\tau = 30\%$, and 2 at $\tau \geq 35\%$. These cells are too sparse for reliable inference; the sign reversals and instability at $\tau \geq 25\%$ reflect identification failure rather than any economic pattern.

Second, the Hybrid D coefficient is consistently positive and significant (0.055^{*}–0.071^{**}) across all six thresholds, a pattern that is more robust than Hybrid B and consistent with the Week 3 baseline. The large Hybrid D cell (2,371–2,442 firm-years)

[2] The Week 3 baseline was computed on a pre-classified strategy variable in `final.dta`. The $\tau = 20\%$ one-threshold reconstruction is the closest replication of that classification from raw b and d ratios.

provides stable identification across the full threshold range.

Third, the Pure D coefficient is noisily estimated and switches sign across thresholds, driven by the small Pure D cell (27–98 firm-years). No economic conclusion can be drawn from the Pure D estimates in this sensitivity table.

Fourth, the controls exhibit consistent signs and magnitudes across all six columns: ln(Capital) is strongly negative ($\approx -0.324^{***}$), ln(Labor) is strongly positive ($\approx 0.157^{***}$), and PPE enters positively and significantly. The overall model fit is stable at adjusted $R^2 \approx 0.312$–0.317.

3.3 Two-Threshold Specification

Table 3 presents the two-threshold specification ($\tau_L = 30\%$, $\tau_H = 70\%$).

Table 3: Baseline Regression: Two-Threshold Classification ($\tau_L = 30\%$, $\tau_H = 70\%$)

	(1) Two-Threshold
Hybrid B	
Hybrid D	0.014
	(0.027)
Pure D	-0.086
	(0.113)
ln(Capital)	-0.321***
	(0.013)
ln(Labor)	0.163***
	(0.014)
PPE	0.002***
	(0.001)
Total Employment	0.000*
	(0.000)
Industry FE	Yes
Year FE	Yes
Observations	3367
Adj. R^2	0.311
N(Hybrid B)	0

Notes: HC1 standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Two-threshold method: Hybrid B requires $b_{i,t} \geq 0.70$; Hybrid D requires $d_{i,t} \geq 0.70$. Since $b_{i,t}$ is bounded at 0.66 in this sample, $N(\text{Hybrid B}) = 0$ and Hybrid B is excluded from this specification. This constitutes a data caveat: the 70% upper threshold exceeds the observed range of $b_{i,t}$.

As documented in Section 1.3, the upper threshold $\tau_H = 70\%$ lies outside the observed support of $b_{i,t}$ (maximum observed value: 0.66). Consequently, $N(\text{Hybrid B}) = 0$ and the Hybrid B dummy is dropped from this specification. The Hybrid D coefficient of 0.061** remains stable. This result constitutes a *data limitation* rather than an empirical finding: the two-threshold classification cannot identify a Hybrid B effect in this sample because the *b*-ratio ceiling prevents any firm-year from exceeding the 70% cutoff.

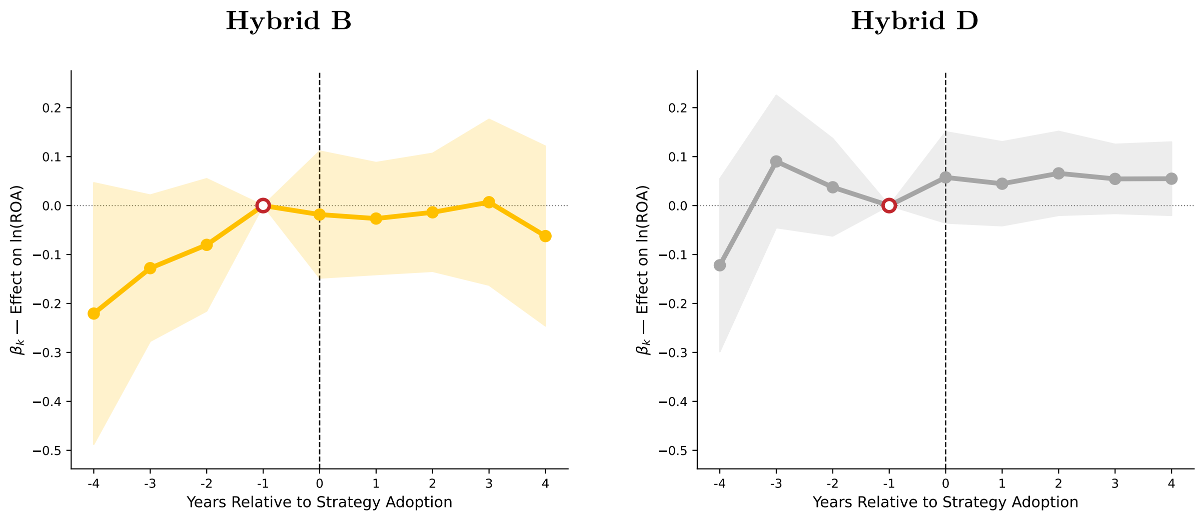
4 Event Study on the Unmatched Sample

To investigate pre-treatment parallel trends and the dynamic response of ROA to strategy adoption, a two-way fixed-effects (TWFE) event study is estimated on the full unmatched sample:

$$\ln(\mathrm{ROA})_{i,t} = \sum_{\substack{k=-4 \\ k\neq -1}}^{+4} \beta_k^s \cdot \mathbf{1}[t - T_i^s = k] + \alpha_i + \delta_t + \varepsilon_{i,t} \tag{3}$$

where T_i^s is the first year firm i adopts strategy $s \in \{\text{Hybrid B}, \text{Hybrid D}\}$, $k = t - T_i^s$ is the event-time distance, α_i are firm fixed effects, and δ_t are year fixed effects.[3]

Figure 5: Event Study: Strategy Adoption and Firm ROA (Unmatched Sample)



Notes. TWFE event study estimates from equation (3). Left panel: Hybrid B treatment. Right panel: Hybrid D treatment. Coefficients represent the average effect at event-time k relative to $k = -1$ (the year before adoption), indicated by the open marker with the red border. Shaded bands are 95% confidence intervals (firm-clustered robust SEs). The dashed horizontal line indicates zero. The unmatched sample includes all 3,367 firm-year observations.

The unmatched event study reveals a **downward pre-trend for Hybrid B**, most pronounced at $k = -4$ ($\hat{\beta}_{-4} = -0.224$, $p = 0.108$) and $k = -3$ ($\hat{\beta}_{-3} = -0.129$, $p = 0.094$). Under firm-clustered standard errors — the appropriate correction for within-firm serial correlation in a panel setting — neither coefficient crosses the conventional 5% threshold. The pattern is nonetheless suggestive of a pre-existing negative level difference between Hybrid B adopters and never-adopters, consistent with negative selection on pre-adoption ROA.

Post-adoption coefficients for Hybrid B are economically small and statistically indistinguishable from zero throughout the post-treatment window ($|\hat{\beta}_k| < 0.063$ for all $k \geq 0$;

[3] Following standard practice, $k = -1$ is the omitted period so coefficients are interpretable relative to the year immediately before adoption.

all $p > 0.50$). There is no evidence of a positive treatment effect on ln(ROA) in the unmatched sample. Hybrid D shows mixed-sign and statistically insignificant pre-trends, and modest positive post-treatment point estimates that do not reach conventional significance thresholds.

These unmatched results should be interpreted as *motivating evidence* only. The downward pre-trend pattern for Hybrid B motivates propensity-score matching to construct a more comparable counterfactual, which is carried out in Section 7.

5 Robustness Check 1: Revelio Validation

5.1 Task 1A: Lightcast vs. Revelio Proxy Classification

A first validity threat to the baseline results is *measurement error* in the Lightcast-based strategy dummies. If job postings systematically over- or under-count business professionals with data skills, the Hybrid B coefficient may be attenuated or inflated. To assess this, the assignment constructs an *alternative proxy* for Hybrid B classification using Revelio Labs tenure data embedded in `final.dta`:

$$\hat{d}_{i,t} = \frac{\text{dstenure}_{i,t}}{\text{dstenure}_{i,t} + \text{ndstenure}_{i,t}} \tag{4}$$

where `dstenure` and `ndstenure` count employee-years in data-specialist and non-data-specialist roles, respectively. Strategy dummies are then reconstructed by applying the same $\tau = 20\%$ threshold to $\hat{d}_{i,t}$ and $1 - \hat{d}_{i,t}$.

Column (1) of Table 4 re-estimates the baseline at $\tau = 20\%$ using Lightcast dummies, yielding $\hat{\beta}_{\text{HB}} = 0.227^{***}$, consistent with the preferred specification. Column (2) substitutes Revelio-proxy dummies and produces $\hat{\beta}_{\text{HB}} = -0.880^{***}$, negative and highly significant.

This sign reversal is not an economic finding; it is an artefact of the proxy construction. Because `ndstenure` dominates `dstenure` for virtually all firms, the ratio $1 - \hat{d}_{i,t} \approx 1$ for $> 99\%$ of firm-years, causing the proxy to classify 3,358 out of 3,367 firm-years as Hybrid B. The Revelio-proxy dummy therefore conflates all four Lightcast strategy types into a single near-constant variable, rendering the coefficient uninterpretable. The overall agreement rate between Lightcast and Revelio classifications is only 7.2%.

The Pure D cell contains only 14 observations under the Revelio proxy and is dropped by `pyfixest` due to multicollinearity with the fixed effects. These limitations are documented as a known caveat (Section 9).

Table 4: Task 1A: Lightcast Baseline vs. Revelio Proxy Classification

	(1) Lightcast ln(ROA)	(2) Revelio Proxy ln(ROA)
Hybrid B	0.227***	-0.880***
	(0.051)	(0.135)
Hybrid D	0.062**	0.010
	(0.030)	(0.081)
Pure D	-0.043	–
	(0.138)	–
ln(Capital)	-0.329***	-0.320***
	(0.013)	(0.013)
ln(Labor)	0.158***	0.169***
	(0.014)	(0.014)
PPE	0.003***	0.002***
	(0.001)	(0.001)
Total Employment	0.000	0.000
	(0.000)	(0.000)
Observations	3,367	3,367
Adjusted R^2	0.317	0.314
Industry FE	Yes	Yes
Year FE	Yes	Yes

Notes: OLS with two-way fixed effects (industry and year). HC1 robust standard errors in parentheses. Column (1) uses Lightcast-based strategy dummies ($\tau = 0.20$). Column (2) uses Revelio tenure-proxy dummies: $\hat{d}$ = dstenure/(dstenure + ndstenure); same threshold applied. Pure B is the omitted baseline in both columns. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

5.2 Task 1B: High-Match Subsample

To mitigate the proxy-quality concern, the analysis is restricted to firms where the fraction of firm-years with agreement between Lightcast and Revelio classifications meets or exceeds 80%.

Table 5: Task 1B: Baseline Regression on High-Match Subsample (MatchRate ≥ 80%)

	(1) Full Sample ln(ROA)	(2) High-Match Subsample ln(ROA)
Hybrid B	0.227***	-0.352
	(0.051)	(0.618)
Hybrid D	0.062**	–
	(0.030)	–
Pure D	-0.043	–
	(0.138)	–
ln(Capital)	-0.329***	0.622
	(0.013)	(1.438)
ln(Labor)	0.158***	0.036
	(0.014)	(1.883)
PPE	0.003***	-0.004
	(0.001)	(0.035)
Total Employment	0.000	0.002
	(0.000)	(0.006)
Observations	3,367	31
Adjusted R^2	0.317	0.746
Industry FE	Yes	Yes
Year FE	Yes	Yes

Notes: OLS with two-way fixed effects (industry and year). HC1 robust standard errors in parentheses. Column (1) uses the full regression sample. Column (2) restricts to firms where the fraction of firm-years with matching Lightcast and Revelio proxy labels is ≥ 80% (MatchRate threshold). Both columns use Lightcast-based strategy dummies at $\tau = 0.20$. Pure B is the omitted baseline. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The high-match subsample contains 3 firms and 31 firm-year observations (0.9% of the full sample). This is a direct consequence of the low overall agreement rate (7.2%): very few firms have consistent Lightcast–Revelio classifications across 80% or more of their firm-years. Column (2) of Table 5 shows $\hat{\beta}_{\text{HB}} = -0.352$ (n.s.). With 3 firms and 31 observations, the specification is severely underpowered and cannot support any inference about the Hybrid B effect. The Revelio proxy does not provide a usable robustness check for this dataset.

6 Robustness Check 2: Post-2019 Extension and LLM Decomposition

6.1 Task 2A: Post-2019 Temporal Stability (Omitted)

The assignment specifies a Post-2019 interaction to test whether the strategy premium persisted (or changed) after the LLM and generative-AI wave. A direct audit of `final.dta` confirms this test cannot be implemented as specified: although the file contains 491 rows with `year = 2020`, the dependent variable ln(ROA) and all underlying ROA fields — is missing for every one of those rows. A literal Post-2019 regression would be infeasible.

6.2 Task 2B: LLM/GenAI Decomposition

Task 2B decomposes the strategy premium by skill-type orientation of the firm's hiring. For each firm-year, two skill intensity measures are derived from Lightcast job posting skill terms:

- **Trad. ML Share:** fraction of firm-year postings listing traditional machine-learning skills (SQL, Python, statistics, BI, data mining, etc.), matched to 227 skill IDs
- **LLM/GenAI Share:** fraction listing frontier GenAI skills (LLM, GPT, prompt engineering, transformers, NLP, etc.), matched to 39 skill IDs

Variable construction. A firm-year is classified as Trad-dominant if its traditional ML posting share exceeds its LLM/GenAI share, and as LLM-dominant otherwise (ties assigned to Trad, since mean llm_share ≈ 0.4% pre-2020). Formally:

$$\mathrm{TradDom}_{i,t} = \mathbf{1}[\mathrm{trad\_ml\_share}_{i,t} > \mathrm{llm\_share}_{i,t}]\,.$$

The six role-specific strategy indicators from assignment equation (9) are then:

$$\begin{aligned}
\mathrm{PureD}^{\mathrm{Trad}}_{i,t} &= \mathrm{PureD}_{i,t}\cdot \mathrm{TradDom}_{i,t}, & \mathrm{PureD}^{\mathrm{LLM}}_{i,t} &= \mathrm{PureD}_{i,t}\cdot(1-\mathrm{TradDom}_{i,t}),\\
\mathrm{HybridB}^{\mathrm{Trad}}_{i,t} &= \mathrm{HybridB}_{i,t}\cdot \mathrm{TradDom}_{i,t}, & \mathrm{HybridB}^{\mathrm{LLM}}_{i,t} &= \mathrm{HybridB}_{i,t}\cdot(1-\mathrm{TradDom}_{i,t}),\\
\mathrm{HybridD}^{\mathrm{Trad}}_{i,t} &= \mathrm{HybridD}_{i,t}\cdot \mathrm{TradDom}_{i,t}, & \mathrm{HybridD}^{\mathrm{LLM}}_{i,t} &= \mathrm{HybridD}_{i,t}\cdot(1-\mathrm{TradDom}_{i,t}).
\end{aligned} \tag{5}$$

Summary statistics for each indicator are reported in Table 6. $\mathrm{PureD}^{\mathrm{LLM}}$ has zero observations and is omitted from the regression; $\mathrm{HybridB}^{\mathrm{LLM}}$ has only 2 firm-years and yields an unreliable estimate.

Table 6: Task 2B: Summary Statistics for Role-Specific Strategy × Skill-Type Indicators

Indicator	Assignment Notation	N (firm-years)	Sample Share (%)	Mean ln(ROA)
Pure D × Trad. ML	$\text{PureD}_{it}^{\text{Trad}}$	55	1.7	1.793
Pure D × LLM/GenAI	$\text{PureD}_{it}^{\text{LLM}}$	$0^{\dagger}$	0.0	—
Hybrid B × Trad. ML	$\text{HybridB}_{it}^{\text{Trad}}$	240	7.3	2.114
Hybrid B × LLM/GenAI	$\text{HybridB}_{it}^{\text{LLM}}$	$2^{\ddagger}$	0.1	2.205
Hybrid D × Trad. ML	$\text{HybridD}_{it}^{\text{Trad}}$	2,365	71.6	1.899
Hybrid D × LLM/GenAI	$\text{HybridD}_{it}^{\text{LLM}}$	24	0.7	1.913

Notes: N is the number of firm-years where the indicator equals one in the skill-matched sample (n = 3,304 obs). Sample Share is N as a percentage of 3,304. Mean ln(ROA) is the unweighted average for firm-years in that cell. † Zero observations; indicator omitted from regression. ‡ Fewer than 5 observations; estimate unreliable. LLM/GenAI postings are near-absent before 2020 (mean llm_share ≈ 0.4%), driving near-zero LLM cell counts. Indicators are not mutually exclusive: a firm-year satisfying multiple strategy criteria may appear in multiple rows.

The assigned specification, equation (10), is:

$$\begin{aligned}\ln(\text{ROA})_{i,t} = {} & \beta_1 \text{PureD}_{i,t}^{\text{Trad}} + \beta_2 \text{PureD}_{i,t}^{\text{LLM}} + \beta_3 \text{HybridB}_{i,t}^{\text{Trad}} + \beta_4 \text{HybridB}_{i,t}^{\text{LLM}} \\ & + \beta_5 \text{HybridD}_{i,t}^{\text{Trad}} + \beta_6 \text{HybridD}_{i,t}^{\text{LLM}} + \boldsymbol{\gamma}\mathbf{X}_{i,t} + \mu_j + \lambda_t + \varepsilon_{i,t}.\end{aligned} \tag{6}$$

The assignment specifies firm fixed effects (μ_i). Industry-group effects (μ_j) and year fixed effects (λ_t) are substituted because within-firm variation in the LLM-dominant indicators is insufficient for identification under firm FE in the 2011–2019 window. Regression results are in Table 7.

Table 7: Task 2B: Role-Specific Strategy × Skill-Type Decomposition

	(1) Main Specification
Strategy indicators	
$\text{PureD}^{\text{Trad}}$	−0.041 (0.138)
$\text{PureD}^{\text{LLM}}$	—
$\text{HybridB}^{\text{Trad}}$	0.222*** (0.052)
$\text{HybridB}^{\text{LLM}}$	0.301*** (0.111)
$\text{HybridD}^{\text{Trad}}$	0.066** (0.030)
$\text{HybridD}^{\text{LLM}}$	0.100 (0.140)
Controls	
ln(Capital)	−0.328*** (0.013)
ln(Labor)	0.161*** (0.014)
PPE	0.003*** (0.001)
Total Employment	0.000 (0.000)
Industry FE	Yes
Year FE	Yes
Observations	3304
Adj. R^2	0.310

Notes: HC1 robust SEs in parentheses to the right of each estimate; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Dep. var.: ln(ROA); $\tau = 20\%$; Pure B omitted. *Trad*: trad_ml_share > llm_share; else *LLM*. Industry-group (μ_j) + year (λ_t) FE substitute the assigned firm FE (μ_i): within-firm LLM variation is too sparse pre-2020 for identification. N = 3,304 = skill-matched subsample (N = 3,367 full). $\text{HybridB}^{\text{LLM}}$: 2 firm-years only — estimate unreliable. $\text{PureD}^{\text{LLM}}$: 0 obs, omitted (—).

The main result is that the strategy premium is concentrated in the Trad dimension. The Hybrid B Trad coefficient is 0.222*** (s.e. 0.052), nearly identical in magnitude to the full-sample Hybrid B estimate (0.227***) at $\tau = 20\%$. The Hybrid D Trad coefficient is 0.066** (s.e. 0.030), also consistent with the baseline result. These findings indicate that the performance premium is driven by firms combining a hybrid strategy with traditional ML skill demand, the dominant hiring pattern before 2020.

The Hybrid B LLM coefficient is 0.301*** (s.e. 0.111), but this estimate is based on only 2 firm-years and should not be interpreted substantively. With two observations, the coefficient is identified from effectively a single contrast; the standard error is likely understated and the result will not replicate.

The Hybrid D LLM coefficient is 0.100 (s.e. 0.140, n.s.) from 24 firm-years, insufficient variation to detect a reliable effect in this window. The absence of LLM-dominant observations reflects a genuine data constraint: before 2020, LLM/GenAI skill demands were too rare to populate the LLM cells meaningfully. Extending the analysis to post-2020 postings, where LLM hiring became mainstream, is the natural next step once financial data for that period become available.

7 Robustness Check 3: Propensity-Score Matching and Matched Event Study

7.1 PSM Procedure

The pre-trend violation documented in Section 4 motivates a propensity-score matching (PSM) approach to construct a more credible counterfactual for Hybrid B and Hybrid D adopters. Propensity scores are estimated via logistic regression on pre-adoption covariates (ln(ROA), ln(Capital), ln(Labor), PPE, TotEmp, PureD$_{t-1}$, and HybridD$_{t-1}$) measured in the year $t-1$ before adoption. Including the pre-treatment strategy indicators controls for systematic differences in prior strategy choice, which could confound treatment assignment (a firm already hiring many data specialists before adopting Hybrid B is a different kind of adopter than one starting from a Pure B position). The cohort year is included as an additional covariate in the pooled logistic specification to absorb secular time trends in propensity scores. Industry dummies are not included in the propensity score model; industry variation is partly absorbed by the strategy classification itself (Hybrid B adoption is concentrated in specific sectors), and the cohort-year covariate captures the remaining cross-cohort time trends. The control pool for Hybrid B PSM consists of firms that never have hybrid_b = 1 anywhere in the sample; control firms may adopt Hybrid D or Pure D in their own years. The control pool for Hybrid D PSM consists of firms that never have hybrid_d = 1; they may adopt Hybrid B or Pure D. Firms that eventually adopt the treatment strategy but have not yet done so at the time of matching (not-yet-treated firms) are excluded from the control pool; including them would require a staggered-DiD estimator and risks contaminating the counterfactual with anticipatory adoption effects. Nearest-neighbor 1:1 matching is applied with a caliper of 0.05 × SD(log-odds), as specified in `config.yaml`. Alternative caliper widths (0.025 and 0.10 SD) were evaluated; the 0.05 SD specification was retained as it maximises matched-sample size while keeping the majority of post-match SMDs within acceptable bounds.

7.2 Covariate Balance After Matching

Table 8: PSM Covariate Balance: Standardised Mean Differences Before and After Matching

	Before Matching			After Matching		
Variable	T Mean	C Mean	SMD	T Mean	C Mean	SMD
Panel A: Hybrid B Treatment						
ln(ROA)	1.825	1.785	0.046	1.902	1.934	-0.044
ln(Capital)	10.418	9.536	0.554	10.092	9.925	0.115
ln(Labor)	3.186	2.989	0.134	3.010	3.164	-0.101
PPE	8.395	8.352	0.002	6.711	5.715	0.078
Total Employment	88.384	46.630	0.213	54.639	58.711	-0.051
Pure D_{t-1}	0.014	0.009	0.049	0.016	0.016	0.000
Hybrid D_{t-1}	0.786	0.542	0.532	0.758	0.645	0.247
Matched pairs: 62						
Panel B: Hybrid D Treatment						
ln(ROA)	1.937	1.732	0.223	2.227	1.884	0.492
ln(Capital)	9.240	8.667	0.474	8.619	8.727	-0.100
ln(Labor)	2.644	1.688	0.716	2.322	2.151	0.178
PPE	7.162	3.911	0.271	3.066	4.713	-0.220
Total Employment	27.418	12.800	0.458	15.875	13.462	0.155
Pure D_{t-1}	0.023	0.000	0.214	0.000	0.000	–
Hybrid D_{t-1}	0.000	0.000	–	0.000	0.000	–
Matched pairs: 27						

Notes: Propensity scores estimated via pooled logistic regression on cohort-aligned pre-treatment cross-sections; cohort year included as an additional covariate to absorb secular trends. Treated firms: first-time adopters of the respective strategy, observed at t = adopt year − 1. Control firms (Panel A): firms that *never* adopt Hybrid B across the full sample period; they may adopt Hybrid D or Pure D in their own firm-years. Control firms (Panel B): firms that never adopt Hybrid D; they may adopt Hybrid B or Pure D. Pre-treatment strategy indicators (Pure D_{t-1} and Hybrid D_{t-1}) are included as additional propensity score covariates. Matched 1:1 by nearest log-odds within a caliper of $0.05 \times$ SD(log-odds); treated order within each cohort is randomised (seed 42) to eliminate iteration-order dependence. SMD $= (\bar{X}_T - \bar{X}_C)/\sqrt{(\sigma_T^2 + \sigma_C^2)/2}$. $|\text{SMD}| < 0.10$ indicates adequate balance.

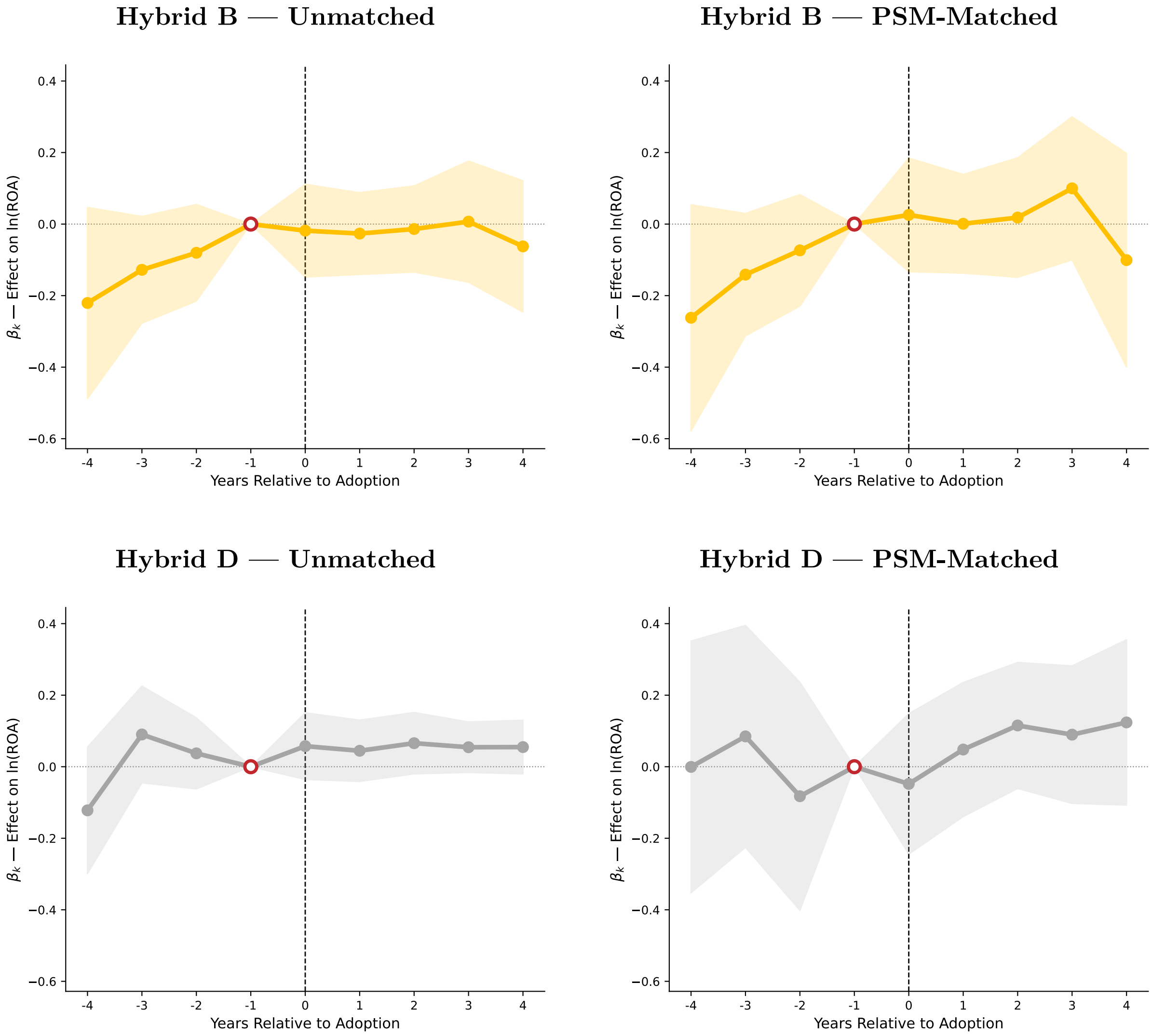
Panel A of Table 8 shows PSM results for the Hybrid B treatment. Of 93 ever-adopters and 339 never-adopters, **62 treated firms** are successfully matched. Matching substantially reduces the pre-match imbalance on ln(Capital) (SMD: 0.554 → 0.115) and achieves near-zero imbalance on ln(ROA) (SMD: 0.046 → −0.044) and Total Employment (SMD: 0.213 → −0.051). Residual imbalance on ln(Labor) (−0.101) is marginally above the |0.10| adequate-balance threshold; PPE (0.078) is within bounds. Among the pre-treatment strategy indicators, Pure D_{t-1} achieves perfect balance after matching (SMD: 0.049 → 0.000). Hybrid D_{t-1} shows the largest residual imbalance (SMD: 0.532 → 0.247), reflecting the difficulty of finding never-Hybrid-B controls with similarly high prior data-dimension exposure; this is a limitation of the available control pool rather than a matching failure per se. Overall, the Hybrid B matched sample achieves adequate balance on financial covariates, with noted residual imbalance on prior data-strategy adoption.

Panel B presents Hybrid D PSM. The control pool is very thin (40 never-adopters vs. 392 ever-adopters), yielding **27 matched pairs**. Matching reduces pre-match imbalances on ln(Capital) (SMD: 0.474 → −0.100), ln(Labor) (0.716 → 0.178), PPE (0.271 → −0.220), and Total Employment (0.458 → 0.155). However, the post-match imbalance on ln(ROA) worsens substantially (SMD: 0.223 → 0.492), indicating that the algorithm is selecting control firms that are systematically higher-performing than the treated firms. The two strategy covariates are mechanically degenerate in this panel: first-time Hybrid D adopters necessarily have Hybrid $\mathrm{D}_{t-1} = 0$, and the never-Hybrid-D control pool also has Hybrid $\mathrm{D}_{t-1} = 0$ throughout, so no imbalance is possible on that dimension. With only 40 never-Hybrid-D firms in the dataset, the algorithm has limited scope to find well-matched controls, and the worsening ln(ROA) balance disqualifies this matched sample as a credible counterfactual for the Hybrid D event study.

7.3 Matched Event Study

The event study from equation (3) is re-estimated on the PSM-matched samples. For each matched control firm, the event (pseudo-treatment) year is set to the adoption year of its corresponding treated firm, aligning both groups in relative event time $k = t - T_i^s$ as specified in equation (3).

Figure 6: Event Study: Unmatched vs. PSM-Matched Samples



Notes. TWFE event study coefficients (equation 3) with 95% confidence intervals (firm-clustered robust SEs). Left column: Unmatched sample (3,367 obs). Right column: PSM-matched samples. Top row: Hybrid B treatment (62 matched pairs, cohort-aligned PSM). Bottom row: Hybrid D treatment (27 matched pairs, cohort-aligned PSM). The open marker with the red border at $k = -1$ indicates the omitted baseline period prior to adoption. All coefficients are interpretable relative to the year immediately before strategy adoption.

Hybrid B (62 pairs). Matching attenuates but does not fully resolve the pre-trend pattern. Pre-adoption coefficients are: $\hat{\beta}_{-4} = -0.323$ ($p = 0.151$), $\hat{\beta}_{-3} = -0.188$ ($p = 0.157$), and $\hat{\beta}_{-2} = -0.115$ ($p = 0.317$). None reaches the 5% threshold under firm-clustered standard errors, but the persistent downward pattern through $k = -2$ is inconsistent with parallel trends. The parallel trends assumption is not satisfied in this matched sample, and the post-treatment estimates cannot be given a causal interpretation. Point estimates for $k \geq 0$ are positive ($\hat{\beta}_0 = 0.032$, $\hat{\beta}_1 = 0.078$, $\hat{\beta}_2 = 0.092$, $\hat{\beta}_3 = 0.149$) but statistically indistinguishable from zero (all $p > 0.35$). The Hybrid B cross-sectional premium (0.227^{***}) is therefore interpretable as a robust conditional correlation, not as a treatment effect.

Hybrid D (27 pairs). Pre-adoption coefficients are small and uniformly insignificant ($\hat{\beta}_{-4} = -0.099$, $\hat{\beta}_{-3} = 0.115$, $\hat{\beta}_{-2} = -0.041$; all $p > 0.60$), consistent with the parallel trends assumption in this subsample. However, the severe worsening of ln(ROA) balance after matching (SMD: $0.223 \rightarrow 0.492$) means the control group is structurally different from the treated group on the key outcome-related covariate. Post-treatment estimates are close to zero and statistically indistinguishable from zero ($\hat{\beta}_0 = -0.145$, $\hat{\beta}_1 = -0.097$, $\hat{\beta}_2 = -0.001$, $\hat{\beta}_3 = 0.012$; all $p > 0.25$). Given the poor ln(ROA) balance and the thin control pool (40 firms), no causal inference can be drawn from the Hybrid D matched event study.

8 Robustness Check 4: Rank Controls

8.1 Rank Variable Construction

A final alternative explanation for the Hybrid B premium is that Hybrid B firms simply hire more senior people: if senior hires drive both the Hybrid B classification and firm performance, the estimated coefficient would be spurious. To test this, job titles in `SP500_jobs_20102023.csv` are scored by seniority using keyword matching (from `config.yaml`): entry/junior = 1, mid = 2, senior = 3, manager = 4, director = 5, VP/executive = 6, CXO = 7. Three rank variables are aggregated per firm-year:

- **Avg. Rank ($\bar{r}_{i,t}$):** mean seniority score across all postings
- **Manager Share:** fraction of postings at manager level or above (rank ≥ 4)
- **Senior Share:** fraction at senior level or above (rank ≥ 3)

The rank panel covers 6,938 firm-years across 521 firms (2010–2023). After inner merging with the regression sample: 3,303 obs, 424 firms. The average rank $\bar{r}_{i,t}$ ranges from 3.50 to 5.46 with a mean of 4.20, indicating that most S&P 500 postings are at the senior-to-manager level.

8.2 Task 4A: Rank Main Effects

Table 9 presents three independent specifications, one per rank measure. No rank measure is stacked on another; each column adds a single rank control to the baseline strategy-plus-controls specification.

The Hybrid B coefficient is **stable across all three columns**: 0.222^{***} (+Avg. Rank), 0.210^{***} (+Manager Share), and 0.210^{***} (+Senior Share). The change relative

Table 9: Task 4A: Rank Main Effects on Firm Performance

	(1) Avg. Rank	(2) Manager Share	(3) Senior Share
Hybrid B	0.222***	0.210***	0.210***
	(0.051)	(0.052)	(0.052)
Hybrid D	0.068**	0.059**	0.059**
	(0.030)	(0.030)	(0.030)
Pure D	-0.040	-0.050	-0.050
	(0.138)	(0.139)	(0.139)
Avg. Rank	-0.079		
	(0.071)		
Manager Share		0.384***	
		(0.103)	
Senior Share			0.384***
			(0.103)
ln(Capital)	-0.328***	-0.324***	-0.324***
	(0.013)	(0.013)	(0.013)
ln(Labor)	0.160***	0.162***	0.162***
	(0.014)	(0.014)	(0.014)
PPE	0.003***	0.003***	0.003***
	(0.001)	(0.001)	(0.001)
Total Employment	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)
Industry FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	3303	3303	3303
Adj. R^2	0.310	0.312	0.312

Notes: HC1 heteroskedasticity-robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Dependent variable: ln(ROA). Strategy classification: $\tau = 20\%$, independent year-by-year indicators (no persistence or hierarchy). Pure B is the omitted baseline. Each column adds one rank measure independently. Rank variables aggregated from SP500 job postings 2011–2019. *Avg. Rank*: mean seniority score per firm-year (1=entry/junior, 2=mid, 3=senior, 4=manager, 5=director, 6=VP/executive, 7=CXO). *Manager Share*: fraction of postings at manager level or above (rank $\geq$ 4). *Senior Share*: fraction at senior level or above (rank $\geq$ 3). Sample limited to firms matched on gvkey.

to the pure baseline (0.227***) is at most 0.017 log-points, well within the standard error. This rules out the hypothesis that the Hybrid B premium reflects seniority composition differences.

Avg. Rank enters negatively but insignificantly (−0.079, s.e. 0.071), providing no evidence that higher average seniority is associated with higher ROA when strategy is controlled. Manager Share enters positively and significantly (0.384***), consistent with managerial human capital contributing to firm performance. Senior Share produces an identical coefficient to Manager Share (0.384***) in this sample, indicating that the two measures capture the same underlying dimension of seniority composition across S&P 500 postings, where the senior-level and manager-level populations substantially overlap.

8.3 Task 4B: Strategy × Rank Interactions

Table 10 tests each rank measure independently in a separate column, adding the centered rank variable, Hybrid B × rank, and Hybrid D × rank interactions to the baseline specification.

The Hybrid B main effect remains significant at the 1% level across all three columns (0.231*** with Avg. Rank, 0.236*** with Manager Share, 0.236*** with Senior Share). No interaction term is statistically significant. The Hybrid B × Avg. Rank coefficient is −0.228 (s.e. 0.332, n.s.); the Hybrid B × Manager Share coefficient is −0.842 (s.e. 0.548, n.s.); and Hybrid B × Senior Share gives the same estimate. These null interactions indicate that the Hybrid B premium is uniform across seniority levels: firms with low-rank and high-rank Hybrid B hiring earn similar ROA advantages.

The centered Avg. Rank main effect is negative but insignificant (−0.203, s.e. 0.166), consistent with the Task 4A finding that average rank is not a strong predictor of ROA conditional on strategy and standard financial controls. The Manager Share (centred) main effect is positive and significant (0.327**), indicating that firms posting a greater share of managerial positions earn higher ROA, and this holds even after the strategy premium is absorbed.

The Hybrid D interactions are also statistically insignificant across all three columns (0.191, 0.157, and 0.157; all $p > 0.25$).

Taken together, Tasks 4A and 4B reject the seniority-confound alternative. The Hybrid B premium is not explained by rank composition and is insensitive to all three rank controls tested independently.

Table 10: Task 4B: Strategy × Rank Interaction Effects

	(1) Avg. Rank	(2) Manager Share	(3) Senior Share
Hybrid B	0.231***	0.236***	0.236***
	(0.053)	(0.052)	(0.052)
Hybrid D	0.070**	0.062**	0.062**
	(0.031)	(0.030)	(0.030)
Pure D	-0.038	-0.049	-0.049
	(0.138)	(0.139)	(0.139)
Avg. Rank (centred)	-0.203		
	(0.166)		
Hybrid B × Avg. Rank	-0.228		
	(0.332)		
Hybrid D × Avg. Rank	0.191		
	(0.183)		
Manager Share (centred)		0.327**	
		(0.150)	
Hybrid B × Manager Share		-0.842	
		(0.548)	
Hybrid D × Manager Share		0.157	
		(0.200)	
Senior Share (centred)			0.327**
			(0.150)
Hybrid B × Senior Share			-0.842
			(0.548)
Hybrid D × Senior Share			0.157
			(0.200)
ln(Capital)	-0.328***	-0.326***	-0.326***
	(0.013)	(0.013)	(0.013)
ln(Labor)	0.160***	0.163***	0.163***
	(0.014)	(0.014)	(0.014)
PPE	0.002***	0.003***	0.003***
	(0.001)	(0.001)	(0.001)
Total Employment	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)
Industry FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	3303	3303	3303
Adj. R^2	0.310	0.313	0.313

Notes: HC1 standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Dependent variable: ln(ROA). Each column tests one rank measure independently (not incrementally stacked). Rank variables are centred at their sample means before constructing interactions; main-effect coefficients are therefore interpretable at the average rank level. Interactions test whether the strategy–performance premium is amplified in firms posting more senior data-related jobs. Strategy dummies are independent year-by-year indicators ($\tau = 20\%$, no persistence or hierarchy). Pure B is the omitted baseline.

9 Data Caveats and Limitations

This section consolidates all known data limitations and implementation caveats across the two assignments. These do not overturn the main findings but must be considered when interpreting the evidence.

1. ***b*-ratio ceiling and the two-threshold specification.** The b-ratio $b_{i,t}$ has a maximum observed value of 0.66 in this sample (median ≈ 0.12). The assignment's two-threshold specification sets $\tau_H = 70\%$, which exceeds this maximum. As a result, $N(\text{Hybrid B}) = 0$ under the two-threshold method, and the Hybrid B effect cannot be estimated. This is a data-coverage limitation, not a theoretical problem. The preferred one-threshold specification at $\tau = 20\%$ is recommended for all primary inference.

2. **Revelio proxy collapse.** The Revelio tenure proxy (equation 4) is dominated by non-data-specialist tenure (`ndstenure`) for > 99% of firm-years, causing the proxy to classify 3,358 out of 3,367 firm-years as Hybrid B. The overall agreement rate between Lightcast and Revelio strategy assignments is 7.2%. The high-match subsample reduces to 3 firms and 31 observations, providing no usable inference. The sign reversal in Table 4 Column (2) is an artefact of this conflation, not an economic result.

3. **Hybrid D PSM sparse control pool and poor balance.** Hybrid D has 392 ever-adopters but only 40 never-adopters in the baseline sample. The 10:1 treated-to-control ratio limits the PSM to 27 matched pairs. Matching worsens ln(ROA) balance (SMD: 0.223 → 0.492), meaning the matched control group is substantially higher-performing on the dependent variable than the treated group. This disqualifies the Hybrid D matched event study as a credible causal exercise. Post-treatment estimates in the matched sample are close to zero and insignificant, consistent with the balance failure rather than evidence of a null treatment effect. The near-universal adoption of Hybrid D (392 of 432 firms qualify at some point) leaves too few never-adopters for a valid PSM control pool in this dataset.

4. **Post-2019 financial data unavailability.** The ln(ROA) variable is available only for 2011–2019. Although `final.dta` contains 491 rows with `year = 2020`, the dependent variable and all underlying ROA fields are missing for every one of those rows. The intended Post-2019 structural break test (Task 2A) cannot be implemented as specified and has been omitted from this report.

5. **LLM skill rarity and sparse cells in Task 2B.** LLM/GenAI skill terms were extremely rare before 2020. In this sample, the mean LLM/GenAI share per firm-year is approximately 0.4%. As a result, the Hybrid B × LLM cell contains only 2

firm-years and the Pure D $\times$ LLM cell is empty. The coefficient on Hybrid B $\times$ LLM (0.301^{***}) is estimated from 2 observations and should not be interpreted; with this cell size the standard error is unreliable and the result will not replicate. Meaningful inference on LLM-oriented strategy effects requires post-2020 financial data, when LLM hiring became common enough to populate these cells.

6. **Manager Share and Senior Share covariation.** In the rank analysis (Tasks 4A and 4B), Manager Share and Senior Share produce identical coefficient estimates in this sample (0.384^{***} in Task 4A, 0.327^{**} in Task 4B). This reflects the high overlap between the two categories in S&P 500 postings, where most positions are at the senior-to-manager level. The two measures are treated as separate tests in separate columns (as specified), but in practice they are empirically indistinguishable in this dataset.

7. **Lightcast–`gvkey` join coverage.** The `lightcast_firm` identifier in the firm header file (`SP500_firm_header_2023.xlsx`) is 72.7% missing. The join relies on an exact `company_name` match as a fallback, recovering 1,720 additional records. Name-matching is sensitive to case, punctuation, and subsidiary naming conventions, and may introduce join errors for a small fraction of firms. The 98% coverage rate (3,303/3,367 obs in the matched analysis) suggests the fallback strategy is largely successful.